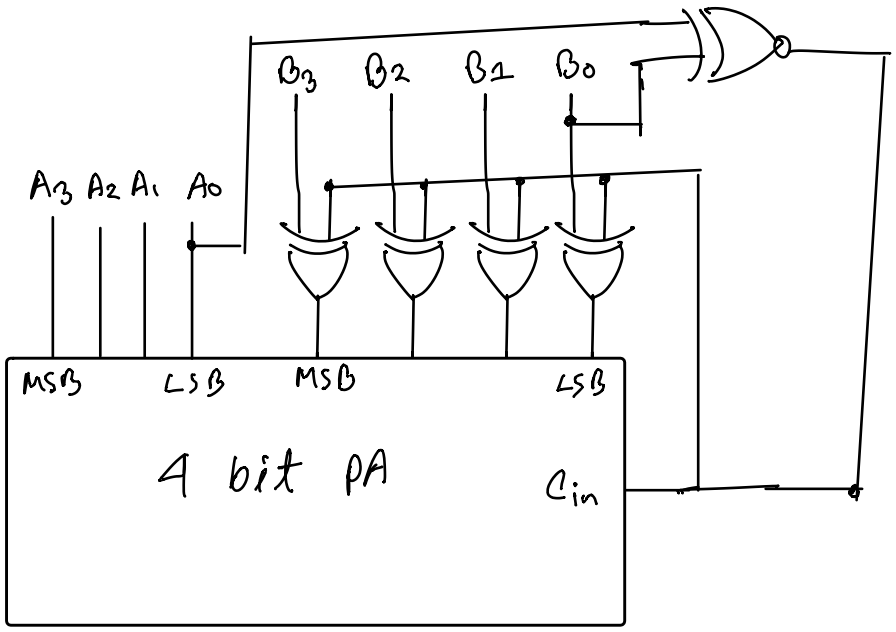
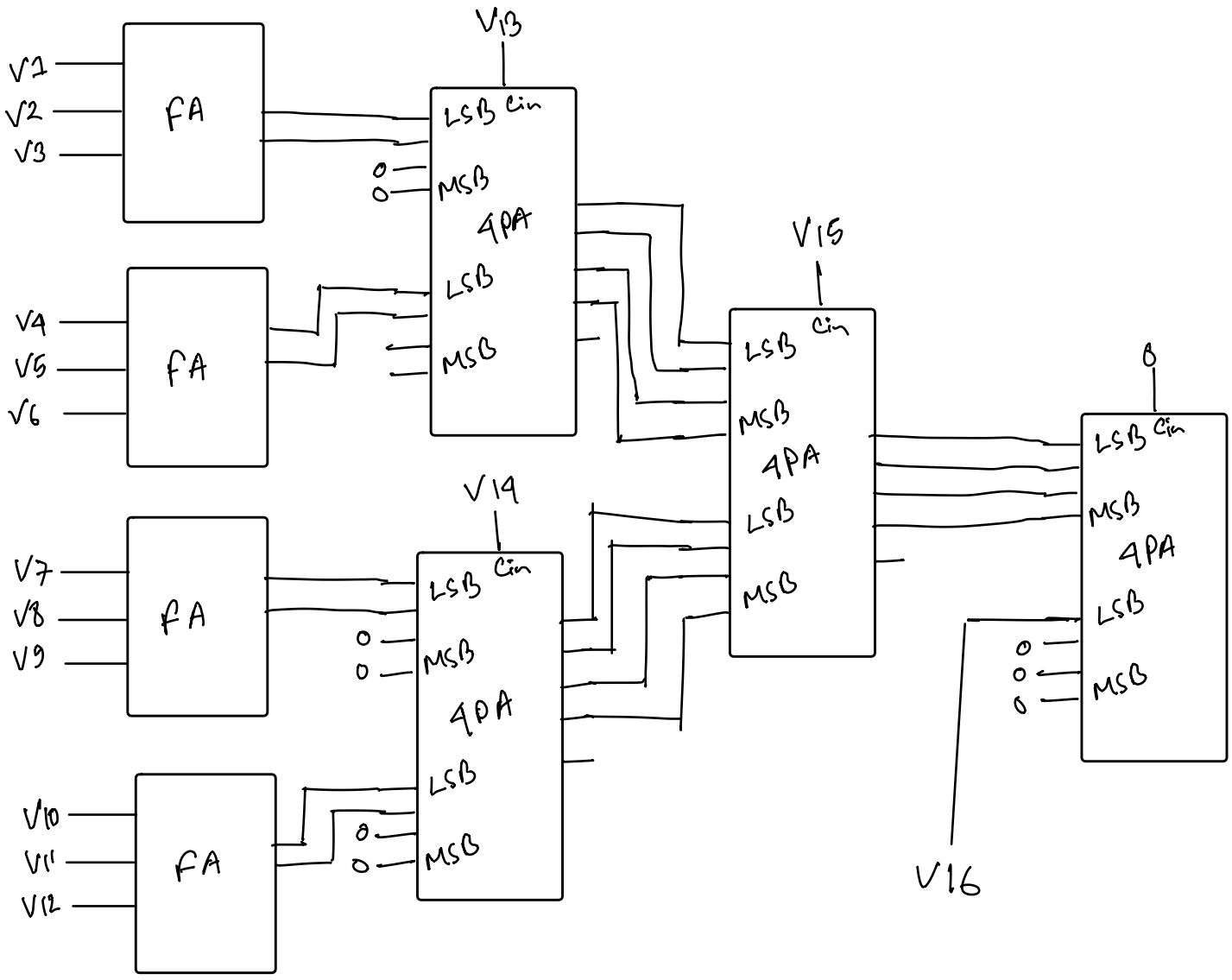
Assignment 3 fall 25

Q1



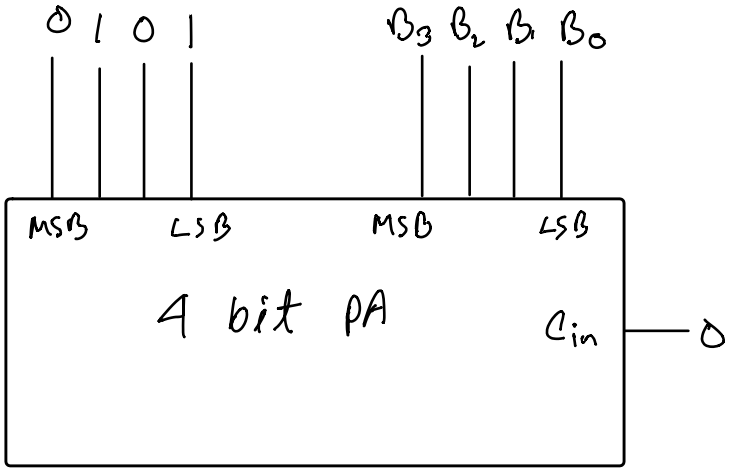
A0	B0	Cin
0	0	1
0	1	0
1	0	0
1	1	1

Q4

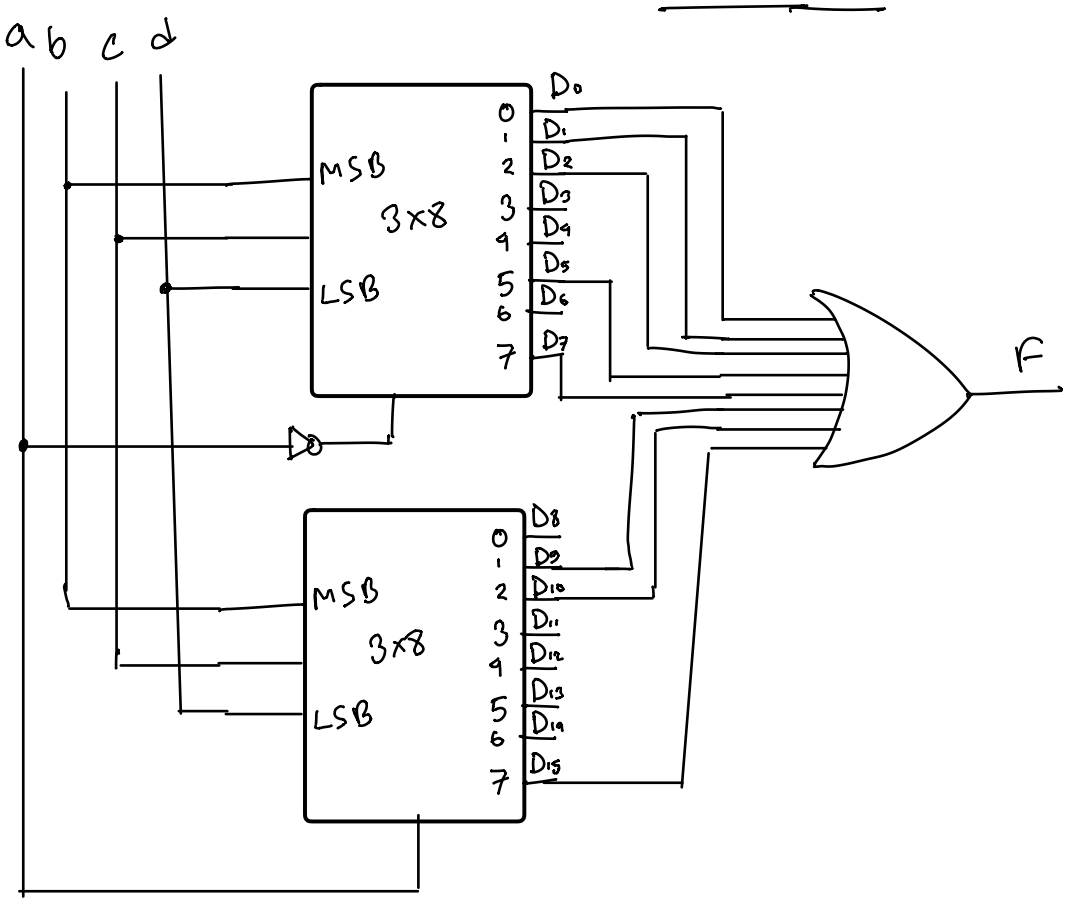


Q5

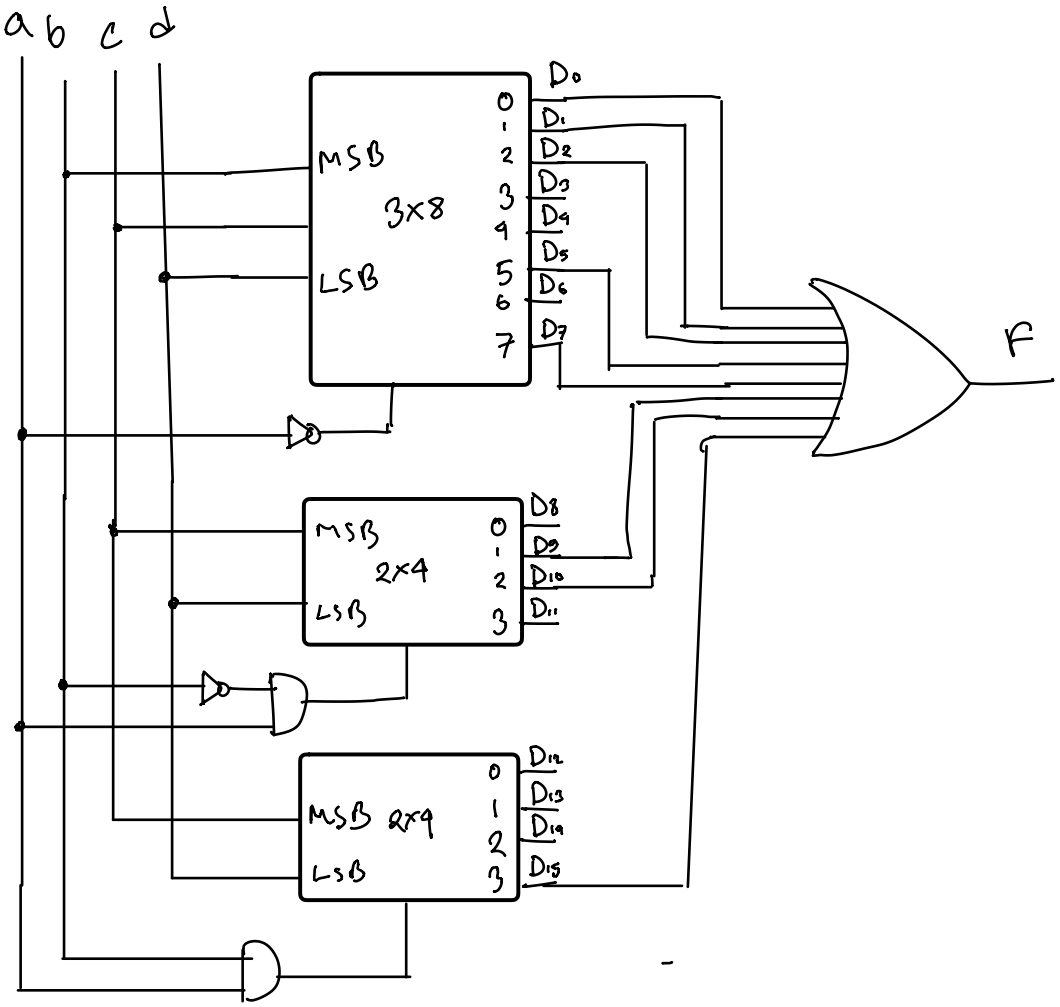
Let BCD values,
$B_3 B_2 B_1 B_0$



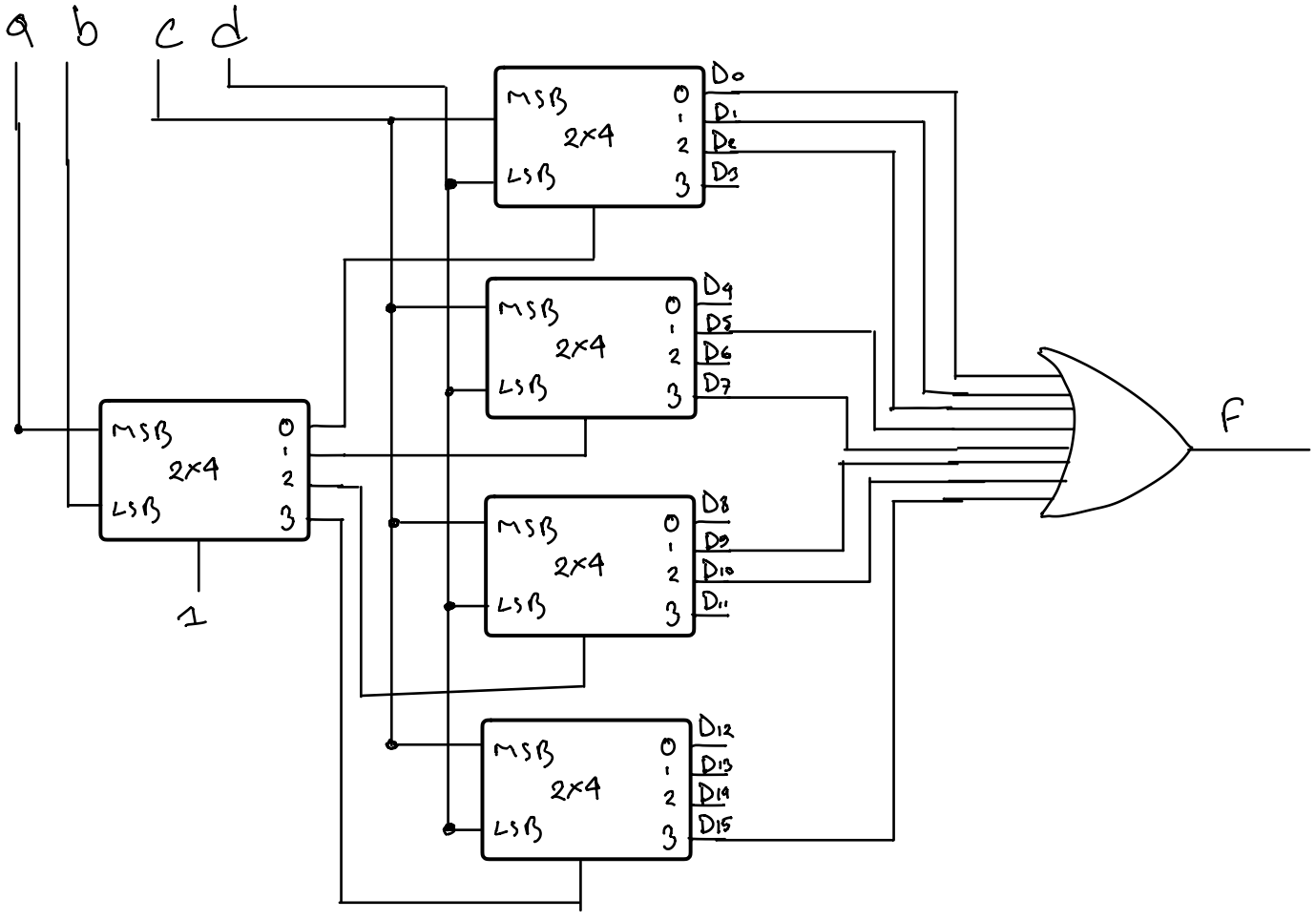
Q6 (a)



Q6(b)


a b c d
MSB
3x8
LSB
0
1
2
3
4
5
6
7
D0
D1
D2
D3
D4
D5
D6
D7
F
MSB
2x4
LSB
0
1
2
3
D8
D9
D10
D11
MSB 2x4
LSB
0
1
2
3
D12
D13
D14
D15

Q6(c)

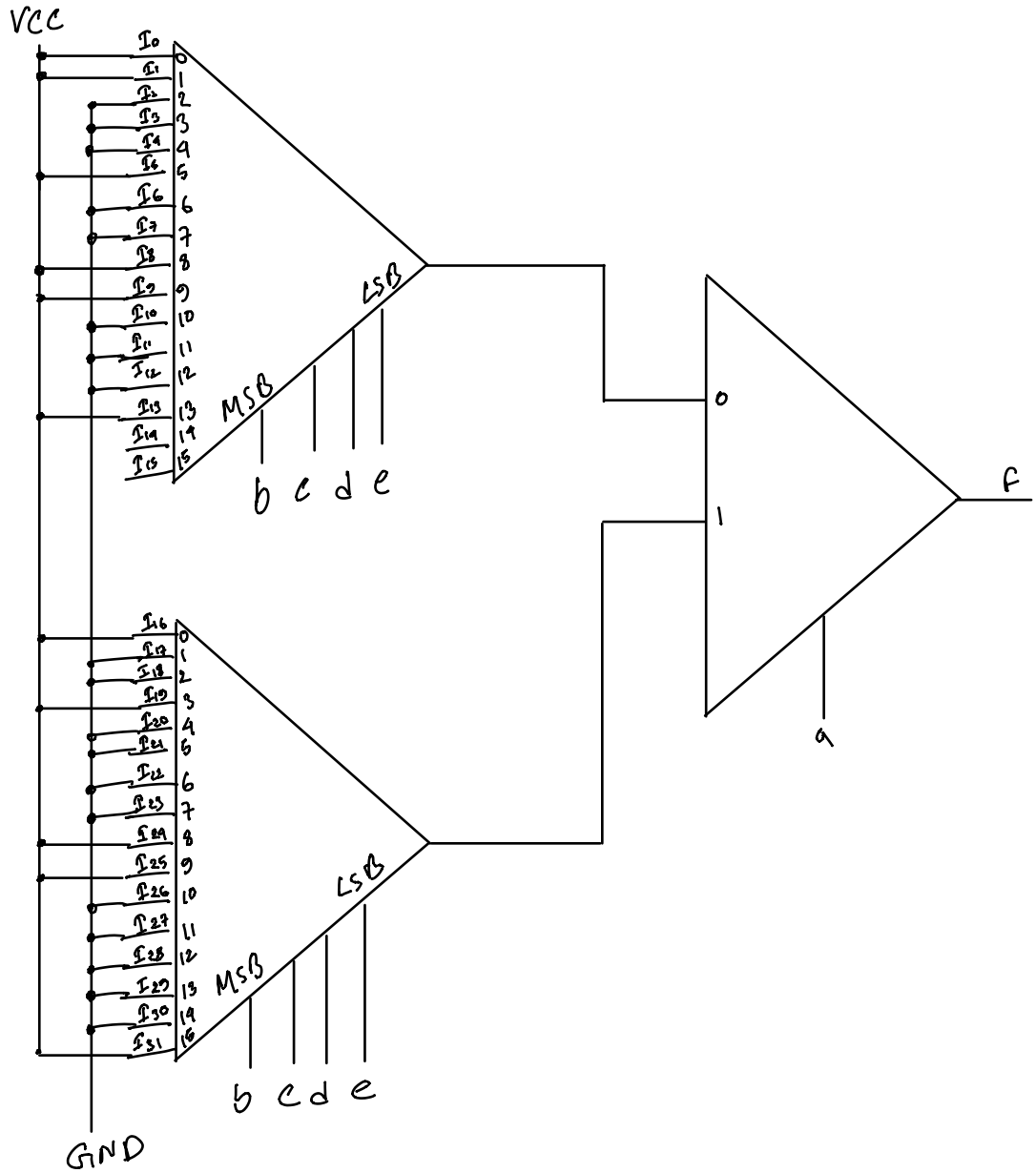

a b c d
MSB
2x4
LSB
0
1
2
3
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
D14
D15
F
1

Q7(a)

We need to build 32×1 Mux using smaller Mux

$f(a, b, c, d, e) = \Sigma(0, 1, 5, 8, 9, 13, 16, 19, 24, 25, 31)$

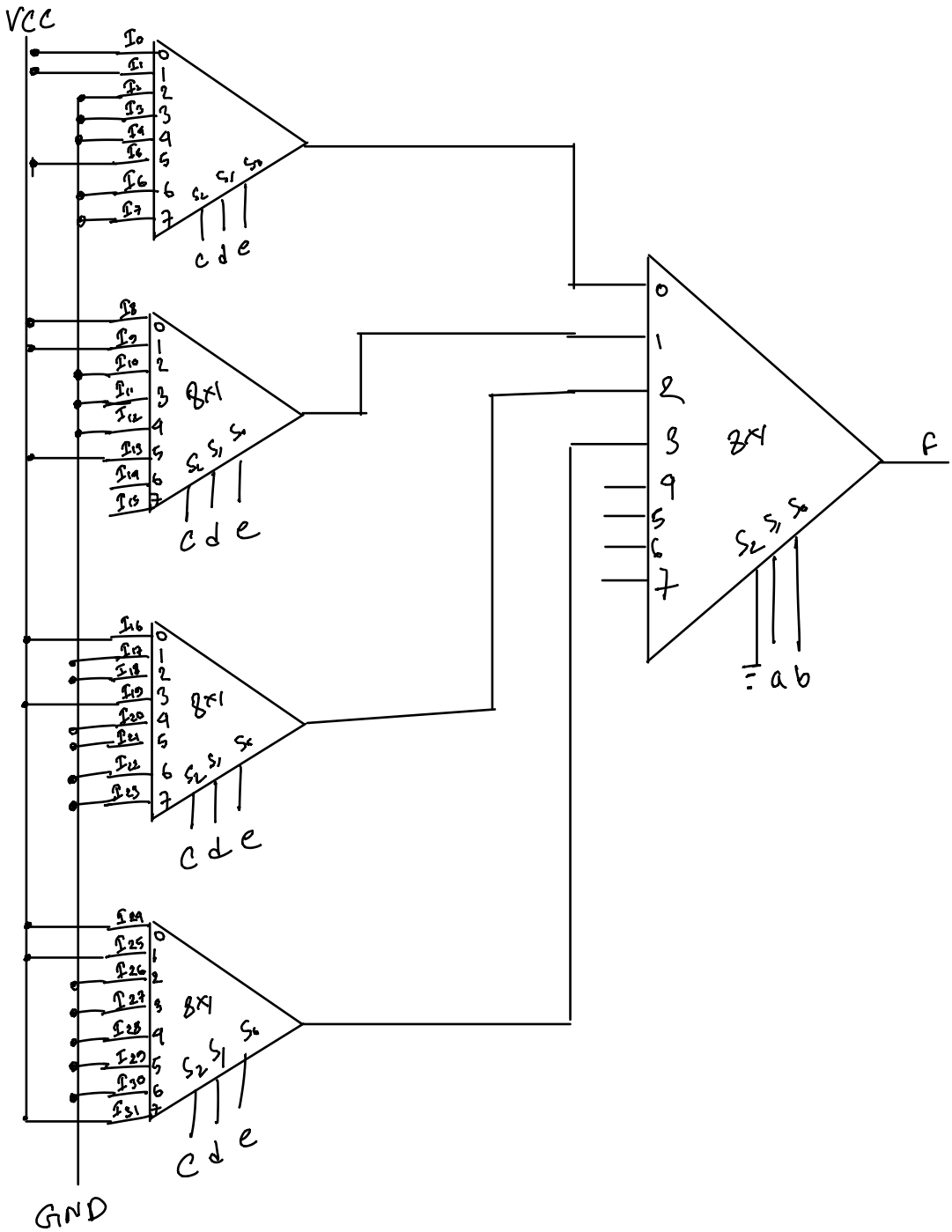
a: MSB, e: LSB



Q7(b)

$$f(a, b, c, d, e) = \Sigma(0, 1, 5, 8, 9, 13, 16, 19, 24, 25, 31)$$

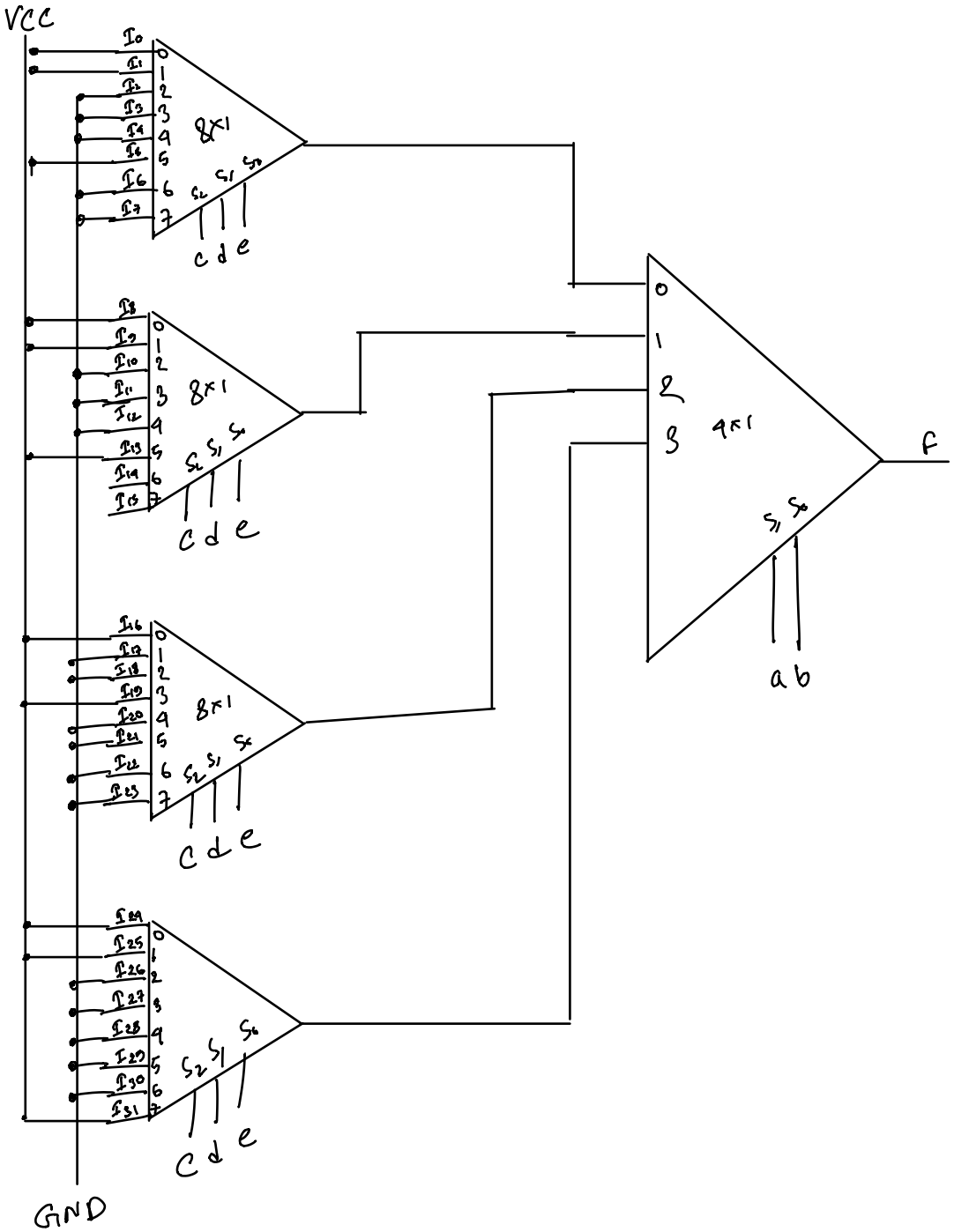
MSB ↑ a, b, c, d, e ↑ LSB



Q7(c)

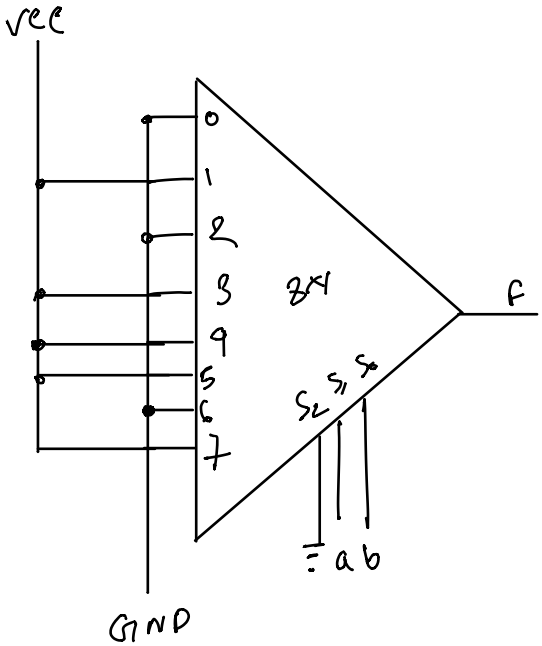
$f(a, b, c, d, e) = \Sigma(0, 1, 5, 8, 9, 13, 16, 19, 24, 25, 31)$

MSB ↑ a, b, c, d, e ↑ LSB



Q8(a)

$F(a,b,c) = \Sigma(1,3,9,5,7)$



Q8(b)

$F(a,b,c,d) = \Sigma(0,2,3,7,8,9,10,11,15)$

	0	1	2	3	4	5	6	7
a'	0	1	2	3	4	5	6	7
a	8	9	10	11	12	13	14	15
	1	0	1	1	0	0	0	1

